Functional Specification

Phantom Inventory Risk Detection

AI-supported POC extending Oracle Retail MFCS with predictive risk scoring and explainability

Version: 1.0
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Document Attribute	Value
Use Case	Phantom Inventory Risk Detection
Primary Source System	Oracle Retail Merchandising Foundation Cloud Service (MFCS)
Historical / Analytical Layer	Oracle Retail AI Foundation / Retail Analytics Platform (AIF/RAP) or equivalent analytical data layer
AI Component	External AI / risk scoring process outside Oracle Retail operational applications
Primary Users	Data analysts, data engineers, QA team, store operations, inventory control and replenishment teams
Primary Output	Ranked list of item/location combinations with phantom inventory risk, explanations and suggested investigation actions

Prepared for delivery, data engineering, analytics and QA teams

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1. Executive Summary

This functional specification defines a proof of concept (POC) for Phantom Inventory Risk Detection. The use case targets discrepancies between system inventory and physical inventory at store level, with the objective of detecting likely phantom stock early, recovering lost sales, improving on-shelf availability and reducing operational losses.

The proposed solution extends Oracle Retail MFCS with a predictive and explainable external AI/risk scoring layer. MFCS remains the operational source of truth for item, location, stock, sales and movement data. AIF/RAP or an equivalent analytical layer is used to provide historical data required for feature engineering, trend detection and model validation.

The POC output is a ranked list of item/location combinations with high phantom inventory risk, including the reason why each case was flagged and a suggested investigation or corrective action.

POC Goal	Expected Outcome
Detect likely phantom inventory	Prioritized item/location risks generated daily or on an agreed batch schedule
Explain risk drivers	Each high-risk record includes top contributing signals and human-readable explanations
Support store validation	Business users can validate flagged records through shelf checks, backroom checks or stock counts
Support QA	QA team receives deterministic formulas, testable acceptance criteria and validation evidence structure
Prepare for production	POC defines data needs, risk logic, feedback loop and implementation roadmap

2. Business Context and Problem Statement

Phantom inventory occurs when the system indicates that inventory is available, but the units are not physically available for sale. This may happen because stock is missing, misplaced, incorrectly received, incorrectly transferred, stolen, damaged, counted incorrectly or affected by transaction/data issues.

The business impact is significant because replenishment may be blocked by apparently positive stock, customers may not find the product, and stores may lose sales while the system continues to report availability.

Business Pain	POC Response
System stock differs from physical stock	Detect positive-stock/no-sales and abnormal stock movement patterns
Lost sales and poor on-shelf availability	Estimate expected demand and potential lost sales value
Manual investigation is not scalable	Generate a ranked list to focus store effort on the highest-risk cases
Users need to trust the AI output	Provide explainable risk drivers and root-cause hypotheses
QA needs testability	Use deterministic formulas and defined validation cases for the POC

3. Scope

3.1 In Scope

- Detection of phantom inventory risk at item/location/business-date level.
- Use of MFCS as the operational source for item, location, stock, sales, movements, counts, receipts, transfers and adjustments where available.
- Use of AIF/RAP or another analytical layer for historical sales, stock and movement history.
- External AI/risk scoring process that calculates risk scores, confidence scores, explanations and suggested actions.
- Ranked output for business validation, store investigation and QA testing.
- Feedback capture from business users to confirm or reject alerts.

3.2 Out of Scope for the POC

- Automatic inventory adjustment posting back into MFCS.
- Automatic replenishment parameter updates.
- Computer vision, shelf camera analysis or image-based validation.
- Real-time intra-day alerting unless the required data is already available.
- Full closed-loop optimization of replenishment, allocation or store labour.
- Fully automated root-cause correction without human validation.

4. Business Objectives and Success Criteria

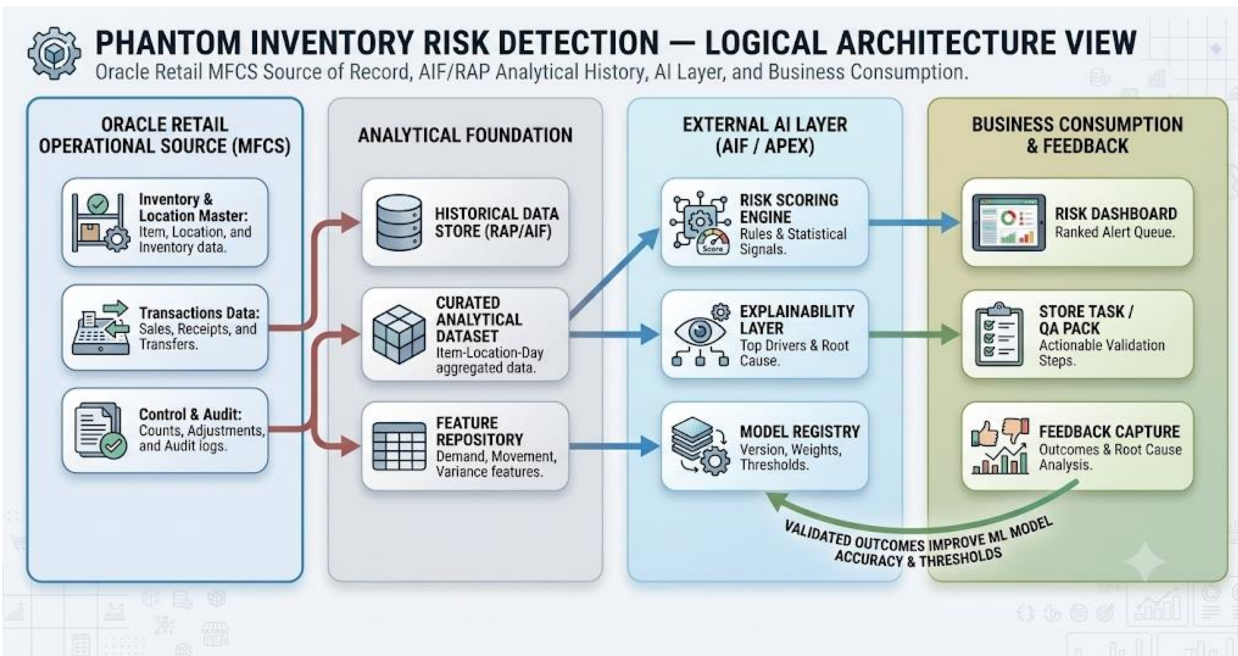
Objective	Success Measure
Detect likely phantom inventory early	High-risk item/location combinations identified before or during business validation
Improve investigation efficiency	Store users receive a short, ranked list instead of broad manual checks
Recover sales opportunity	Flagged cases include estimated lost sales units and value
Improve inventory accuracy	Confirmed cases lead to stock corrections or root-cause actions
Build client confidence	Risk explanations are understandable and aligned with known operational behaviours
Support QA	QA team can reproduce risk calculations and validate output fields against test scenarios

5. Solution Overview

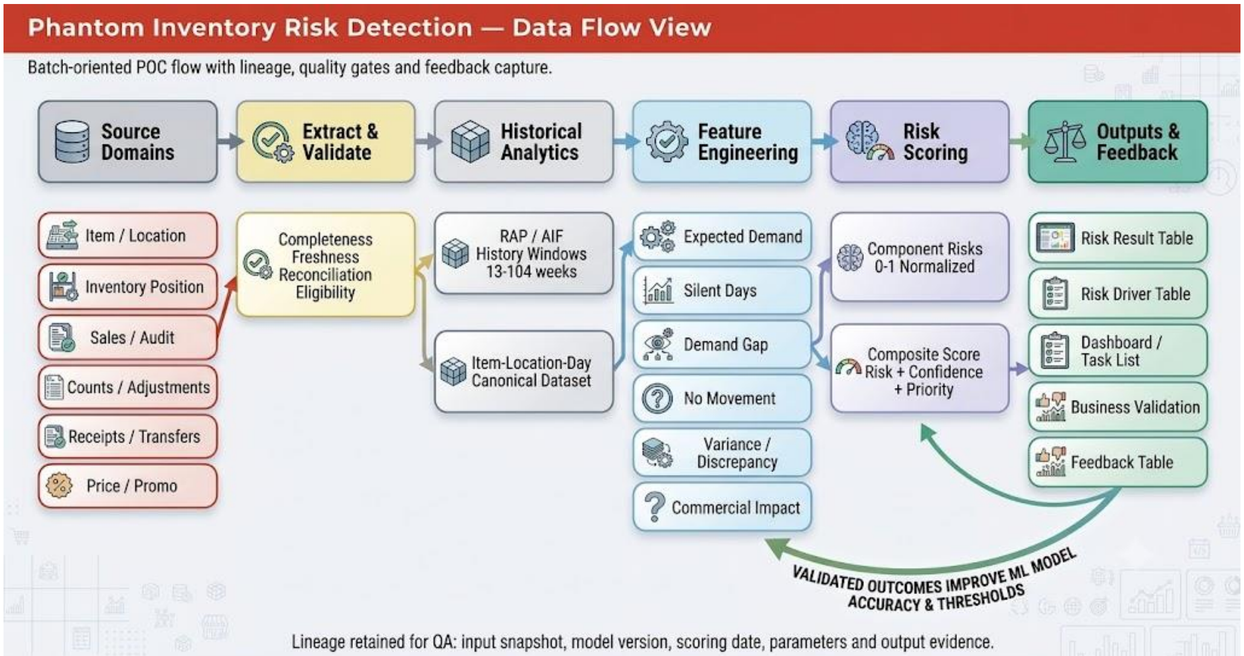
5.1 Logical Architecture

The high-level architecture is shown below:

- `Oracle Retail MFCS`
    - `RAP / AIF Analytical Layer`
    - `External AI / Risk Engine`
    - `Risk Output Layer`
    - `Store Operations, Inventory Control, Replenishment, Data Analysts and QA`

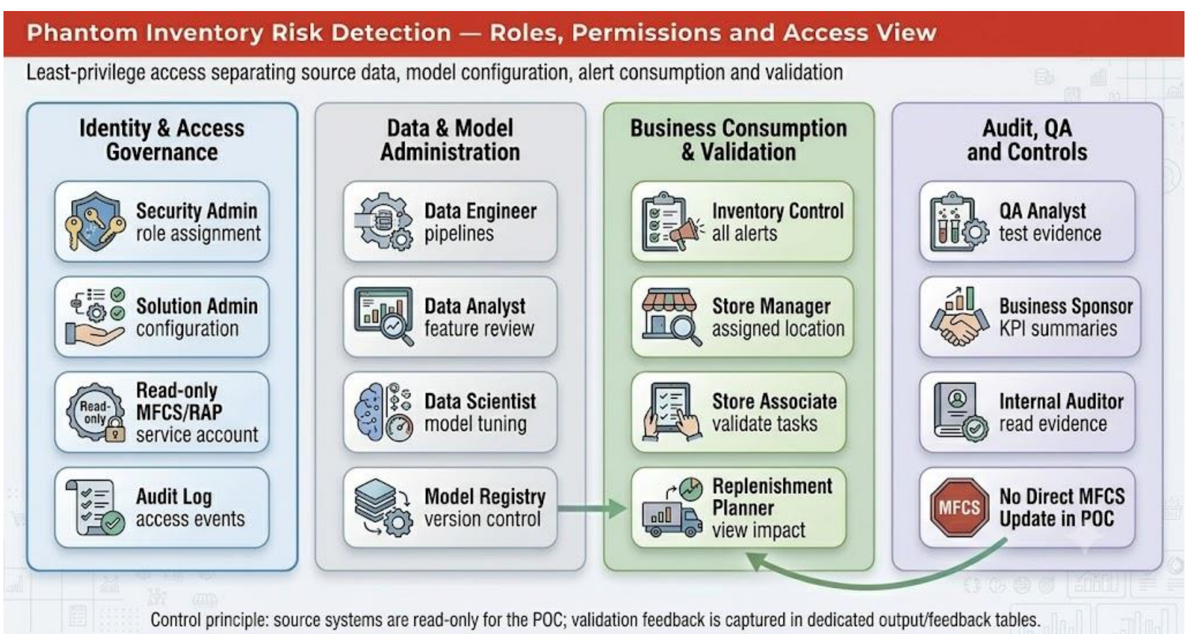


5.2 Data Flow



5.3 Roles, Permissions and User Access

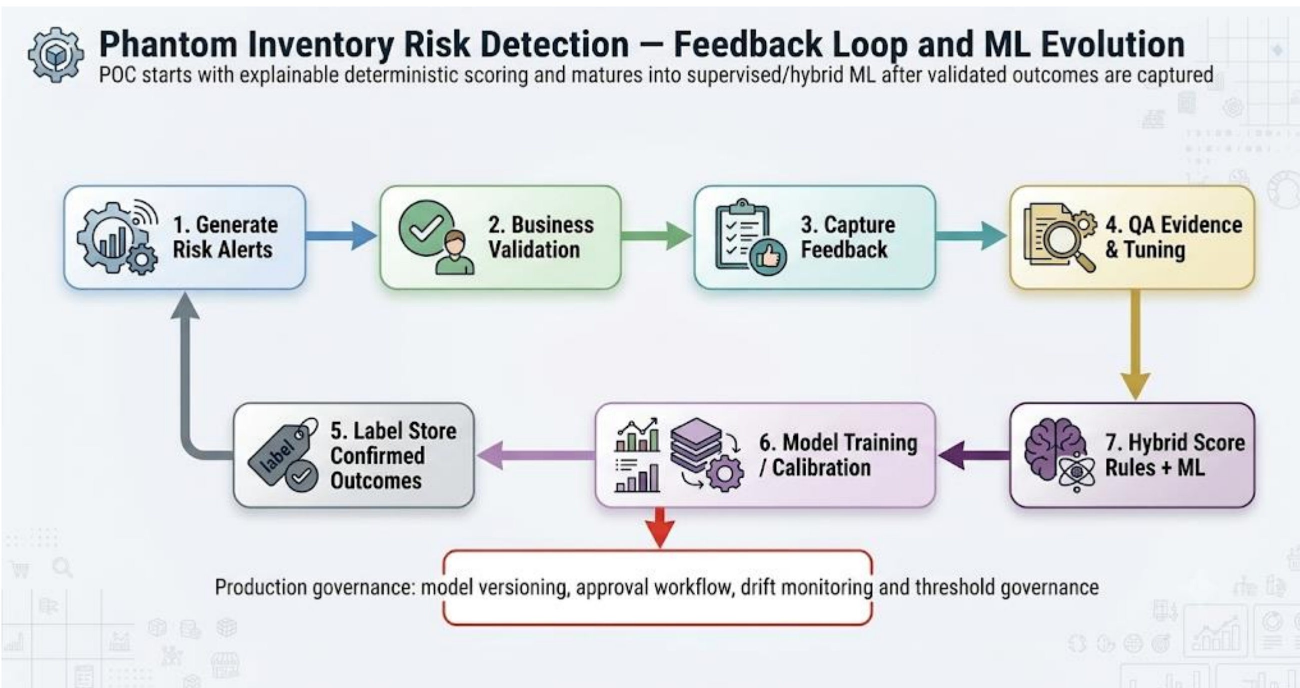
The access view summarizes the role-based operating model and the key POC control that MFCS access remains read-only while validation feedback is captured separately.



5.4 Key Architectural Principles

- MFCS remains the system of record for operational inventory and merchandising data.
- The external AI layer does not directly change inventory in the POC.
- The model output must be explainable and auditable.
- Risk scores should be calculated from reproducible features at item/location/date grain.
- A feedback loop must capture validation outcomes to support QA and future supervised learning.
- Model version, scoring date and input data snapshot date must be stored with every output.

5.5 Feedback Loop and Future ML



6. Main Business Entity and Scoring Grain

The primary scoring entity is:

```
Item + Location + Business Date
```

Each eligible item/location combination receives:

```
Phantom_Inventory_Risk_Score ∈ [0, 100]
```

The solution should also output confidence, priority, top risk drivers, likely root cause and recommended action.

7. Data Requirements

7.1 Source Data from MFCS

Data Domain	Required Data Elements	Purpose
Item master	Item ID, description, hierarchy, status, brand, size, colour, lifecycle status	Product context and eligibility
Location master	Location ID, store status, region, format, cluster, opening/closure indicators	Store context and eligibility
Inventory position	Stock on hand, unavailable stock, reserved stock, damaged stock, available stock if provided	Primary baseline for phantom inventory detection
Sales	Item, location, transaction date, sales units, sales value, transaction status	Demand and no-sales detection
Sales audit	Missing, duplicate, erroneous, suspicious or corrected transaction indicators	Transaction quality and anomaly signal
Stock counts	Book quantity, physical count quantity, count date, variance, count status	Strong validation and feedback signal
Inventory adjustments	Adjustment quantity, reason code, date, user/process source	Shrinkage and correction patterns
Receipts	Expected receipt quantity, actual received quantity, receipt date, PO/ASN references	Receiving discrepancy signal
Transfers	Shipped quantity, received quantity, source/destination location, transfer date	Transfer discrepancy signal
Pricing/promotions	Regular price, selling price, promotion flag, clearance flag	Demand normalization and commercial impact

7.2 Historical Data Requirements

Historical Dataset	Minimum for POC	Preferred for Production	Usage
Daily item/location sales	13 weeks	52–104 weeks	Expected demand and sales suppression
Daily item/location inventory	13 weeks	52–104 weeks	Silent stock and stock cover analysis
Inventory movements	13 weeks	52–104 weeks	No movement and movement anomaly detection
Counts and adjustments	26 weeks	52–104 weeks	Validation, proxy labels and recurrence
Receipts/transfers	13 weeks	52–104 weeks	Recent discrepancy analysis
Pricing/promotions	13 weeks	52–104 weeks	Demand normalization and impact assessment

8. Eligibility and Exclusion Rules

An item/location is eligible for scoring only when the following base conditions are met:

```
Available_Stock > 0
AND Item_Status = 'Active'
AND Location_Status = 'Open'
AND Item_Sellable_At_Location = TRUE
```

Recommended exclusions for the POC:

- Item is new and has insufficient comparable sales history.
- Store is closed or temporarily not trading.
- Item is not ranged or not sellable in the location.
- Item is intentionally blocked from sale, under controlled withdrawal or pending known investigation.
- Data required for scoring is missing or fails critical data quality checks.

9. Risk Signals and Formulas

The POC should use an explainable weighted scoring model. All component risks should be normalized to a value between 0 and 1 before being combined into the final score.

9.1 Available Stock

```
Available_Stock = Stock_On_Hand - Reserved_Quantity - Non_Sellable_Quantity - Damaged_Quantity -
Pending_RTV_Quantity - Customer_Order_Reserved_Quantity

If detailed buckets are unavailable:
Available_Stock = Stock_On_Hand
```

9.2 Expected Demand

```
Expected_Demand = Base_Sales_Rate × Seasonality_Factor × Promotion_Factor × Price_Factor ×
Store_Open_Factor

Base_Sales_Rate = Median daily sales over the last 56 comparable in-stock selling days
```

9.3 Demand Confidence

```
Demand_Confidence = min(1, Expected_Demand_28_Days / Minimum_Demand_Threshold)

Recommended default:
Minimum_Demand_Threshold = 3 units over 28 days
```

9.4 Silent Stock Risk

```
Silent_Days = Consecutive days where Available_Stock > 0 AND Sales_Units = 0

Expected_Demand_Silent_Period = Sum(Expected_Demand) over Silent_Days

Risk_Silent_Stock = 1 - exp(-Expected_Demand_Silent_Period)

Risk_Silent_Stock_Adjusted = Risk_Silent_Stock × Demand_Confidence
```

9.5 Demand Suppression Risk

```
Expected_Sales_Window = Sum(Expected_Demand) over last W days
Actual_Sales_Window = Sum(Sales_Units) over last W days
Recommended W = 14 days

Demand_Gap = max(0, Expected_Sales_Window - Actual_Sales_Window)
Demand_Gap_Ratio = Demand_Gap / max(Expected_Sales_Window, 1)

Risk_Demand_Suppression = min(1, Demand_Gap_Ratio) × Demand_Confidence
```

9.6 No Movement Risk

```
Days_Since_Last_Movement = Business_Date - Date_Of_Last_Movement
Expected_Stock_Cover_Days = Available_Stock / max(Expected_Daily_Demand, 0.01)

Risk_No_Movement = min(1, Days_Since_Last_Movement / max(Expected_Stock_Cover_Days,
Minimum_Movement_Days)) × Demand_Confidence

Recommended Minimum_Movement_Days = 7
```

9.7 Stock Cover Anomaly Risk

```
Stock_Cover_Days = Available_Stock / max(Expected_Daily_Demand, 0.01)
Normal_Cover_Days = Median historical stock cover for item/location or item/store-cluster
Cover_Anomaly_Ratio = Stock_Cover_Days / max(Normal_Cover_Days, 1)

Risk_Stock_Cover_Anomaly = min(1, max(0, Cover_Anomaly_Ratio - 1) / Cover_Anomaly_Tolerance) ×
Demand_Confidence

Recommended Cover_Anomaly_Tolerance = 3
```

9.8 Stock Count Variance Risk

```
Positive_Count_Variance = max(0, Book_Stock_At_Count - Physical_Count_Quantity)
Count_Variance_Ratio = Positive_Count_Variance / max(Book_Stock_At_Count, 1)
Count_Recency_Factor = exp(-Days_Since_Count / Count_Decay_Days)

Risk_Count_Variance = min(1, Count_Variance_Ratio) × Count_Recency_Factor

Recommended Count_Decay_Days = 30
```

9.9 Adjustment History Risk

```
Negative_Adjustment_Units = Sum(abs(Adjustment_Quantity)) where Adjustment_Quantity < 0 over last
56 days
Adjustment_Ratio = Negative_Adjustment_Units / max(Sales_Units_Recent + Available_Stock, 1)

Risk_Adjustment_History = min(1, Adjustment_Ratio / Adjustment_Tolerance)

Recommended Adjustment_Tolerance = 0.20
```

9.10 Receipt Discrepancy Risk

```
Receipt_Discrepancy_Qty = max(0, Expected_Receipt_Qty - Actual_Received_Qty)
Receipt_Discrepancy_Ratio = Receipt_Discrepancy_Qty / max(Expected_Receipt_Qty, 1)

Risk_Receipt_Discrepancy = min(1, Receipt_Discrepancy_Ratio) × exp(-Days_Since_Receipt /
```

```
Receipt_Decay_Days)

Recommended Receipt_Decay_Days = 21
```

9.11 Transfer Discrepancy Risk

```
Transfer_Discrepancy_Qty = max(0, Transfer_Shipped_Qty - Transfer_Received_Qty)
Transfer_Discrepancy_Ratio = Transfer_Discrepancy_Qty / max(Transfer_Shipped_Qty, 1)

Risk_Transfer_Discrepancy = min(1, Transfer_Discrepancy_Ratio) × exp(-Days_Since_Transfer /
Transfer_Decay_Days)

Recommended Transfer_Decay_Days = 21
```

9.12 Historical Recurrence Risk

```
Historical_Phantom_Events = Count of confirmed or proxy phantom inventory events over last 52
weeks

Risk_Historical_Recurrence = min(1, Historical_Phantom_Events / Recurrence_Threshold)

Recommended Recurrence_Threshold = 3
```

10. Composite Risk Score

For the POC, the following transparent weighted model is recommended:

```
Raw_Risk_Score =
  0.25 × Risk_Silent_Stock_Adjusted
+ 0.20 × Risk_Demand_Suppression
+ 0.15 × Risk_No_Movement
+ 0.10 × Risk_Stock_Cover_Anomaly
+ 0.10 × Risk_Count_Variance
+ 0.08 × Risk_Adjustment_History
+ 0.05 × Risk_Receipt_Discrepancy
+ 0.05 × Risk_Transfer_Discrepancy
+ 0.02 × Risk_Historical_Recurrence

Phantom_Inventory_Risk_Score = round(100 × Raw_Risk_Score, 0)
```

Risk Component	Weight	Primary Business Interpretation
Silent stock	25%	Positive stock exists but sales have stopped
Demand suppression	20%	Sales are materially below expected demand
No movement	15%	No sales or inventory movement despite stock availability
Stock cover anomaly	10%	Stock cover is unusually high against expected demand
Stock count variance	10%	Recent count suggests book stock exceeded physical stock
Adjustment history	8%	Recent negative corrections indicate inventory accuracy issue
Receipt discrepancy	5%	Potential receiving issue
Transfer discrepancy	5%	Potential transfer issue
Historical recurrence	2%	Repeated prior phantom patterns

11. Confidence and Commercial Prioritization

11.1 Confidence Score

```
Confidence_Score = Data_Completeness_Score × Demand_Confidence × Model_Stability_Score × 100
```

Confidence should be used to reduce the priority of cases where the risk score is high but the supporting data is weak.

11.2 Estimated Lost Sales

```
Estimated_Lost_Sales_Units = max(0, Expected_Sales_Window - Actual_Sales_Window)

Estimated_Lost_Sales_Value = Estimated_Lost_Sales_Units × Current_Selling_Price
```

11.3 Priority Score

```
Priority_Score = Phantom_Inventory_Risk_Score × Confidence_Score / 100 × Commercial_Impact_Factor

Commercial_Impact_Factor = 1 + Normalized_Estimated_Lost_Sales_Value
```

12. Risk Bands and Action Recommendations

Score Range	Risk Band	Recommended Action
0–39	Low	Monitor only; no immediate action
40–59	Medium	Include in exception report; review if recurring
60–79	High	Request store shelf/backroom check or targeted count
80–100	Critical	Immediate investigation, targeted stock count and root-cause review

13. Explanation and Root Cause Logic

Every high and critical alert must include the top contributing risk drivers and a plain-language explanation.

Likely Root Cause	Typical Trigger Logic	Recommended Action
Possible missing physical stock	High silent stock + high demand suppression	Immediate stock count and inventory investigation
Possible misplaced stock	Positive stock + no sales + no recent receipt/transfer discrepancy	Shelf, fitting room, backroom and display check
Possible receiving issue	Recent receipt discrepancy + no movement after receipt	Review receiving records and physical receiving area
Possible transfer issue	Recent transfer discrepancy + no movement after transfer	Reconcile shipped vs received transfer quantities
Possible shrinkage/theft	Repeated negative adjustments or count write-downs	Inventory control review and targeted count
Possible POS/data issue	Sales audit exceptions or transaction anomalies	Review sales transaction flow and audit exceptions
Possible low-demand false positive	Low demand confidence despite	Monitor only or lower action priority

	positive stock	

13.1 Example Explanation Text

- Positive stock of 8 units has been available for 18 days, but no sales were recorded. Expected demand during this period was 6.3 units.
- Sales are 85% below expected demand over the last 14 days while stock remained positive.
- Recent stock count showed a positive book-to-physical variance of 5 units.
- Stock has not moved for 24 days despite expected daily demand of 0.7 units.
- Recent transfer discrepancy detected: 10 units shipped, 6 units received.

14. Output Data Model

14.1 Main Result Table

Recommended table name: PHANTOM_INVENTORY_RISK_RESULT

Field	Description
alert_id	Unique alert identifier
business_date	Scoring date
item_id	Item identifier
location_id	Store/location identifier
stock_on_hand	Stock on hand from source system or analytical layer
available_stock	Calculated sellable stock
expected_sales_14d	Expected sales over the scoring window
actual_sales_14d	Actual sales over the scoring window
silent_days	Consecutive days with available stock and zero sales
estimated_lost_sales_units	Estimated missed unit sales
estimated_lost_sales_value	Estimated missed sales value
phantom_inventory_risk_score	Score from 0 to 100
confidence_score	Confidence from 0 to 100
priority_score	Operational priority score
risk_band	Low, Medium, High or Critical
likely_root_cause	Main inferred root cause
recommended_action	Suggested business action
model_version	Risk engine version
input_snapshot_timestamp	Timestamp of input data snapshot
created_timestamp	Timestamp of score generation

14.2 Risk Driver Table

Recommended table name: PHANTOM_INVENTORY_RISK_DRIVER

Field	Description
alert_id	Unique alert identifier
business_date	Scoring date
item_id	Item identifier
location_id	Store/location identifier
driver_name	Name of risk signal
driver_score	Normalized signal score between 0 and 1

driver_weight	Applied component weight
weighted_contribution	Contribution to final risk score
explanation_text	Human-readable explanation

14.3 Feedback Table

Recommended table name: PHANTOM_INVENTORY_FEEDBACK

Field	Description
alert_id	Unique alert identifier
item_id	Item identifier
location_id	Store/location identifier
alert_date	Date alert was generated
action_date	Date action was completed
action_type	Shelf check, backroom check, stock count, receiving review, transfer review, ignored
system_stock_qty	System stock before validation
physical_count_qty	Physical quantity found
confirmed_phantom_flag	Yes/No confirmation based on business validation
root_cause_code	Missing, misplaced, shrinkage, receipt issue, transfer issue, data issue, unknown
validated_by	User, role or team validating the case
comments	Optional business comments

15. Key Challenges for the POC

The following challenges should be explicitly managed during the POC because they may affect scoring quality, user trust, QA execution and production readiness.

Challenge	Why It Matters	Mitigation / POC Approach
Data availability across domains	Phantom inventory detection requires sales, stock, movements, counts, receipts, transfers and adjustments. Missing domains reduce signal strength.	Start with mandatory stock/sales signals and progressively add movement/count/receipt/transfer features as available.
Definition of available stock	Different stock buckets may exist for reserved, unavailable, damaged, customer order or pending RTV units.	Agree an explicit available stock calculation with business and MFCS SMEs before scoring.
Low-demand false positives	Slow-moving items can naturally have long periods with no sales.	Use demand confidence and minimum demand thresholds to reduce over-flagging.
Promotion, seasonality and price effects	Sales suppression may be incorrectly inferred if expected demand is not normalized.	Include promotion, price and comparable-day logic where data is available; document limitations where not available.
Historical label scarcity	Confirmed phantom inventory outcomes may not be systematically captured today.	Use explainable rule-based scoring for the POC and capture feedback to create future supervised labels.
Operational validation effort	Store teams cannot validate too many alerts during a POC.	Limit POC output to top-N high-priority cases by store/category and prioritize high confidence/high value

		alerts.
Root-cause ambiguity	A high-risk case may be caused by missing stock, misplaced stock, data issue, receipt issue or transfer issue.	Present likely root cause as a hypothesis, not a definitive conclusion; capture actual root cause during validation.
Data latency	If stock and sales are not synchronized by business date, scoring may generate misleading alerts.	Store input snapshot timestamp and validate data freshness before scoring.
Explainability and user trust	Business users are unlikely to act on black-box scores.	Provide top drivers, formulas, examples and plain-language explanations for every high/critical alert.
QA reproducibility	QA must reproduce model outputs for selected scenarios.	Use deterministic formulas, versioned weights, test datasets and expected-result matrices for validation.
Integration boundaries	The external AI process must not compromise MFCS as the system of record.	Keep POC read-only from MFCS perspective; return outputs to reporting/workflow only.

16. Business User Validation of Risk Scores

Business validation is a required POC activity. It confirms whether the risk scores are meaningful, whether explanations are understandable, and whether recommended actions are operationally practical. This section is also intended to support QA test planning and evidence capture.

16.1 Validation Objectives

- Confirm that high and critical scores correspond to plausible inventory issues.
- Confirm that low and medium scores are not masking obvious high-risk cases.
- Validate that explanations match the data and are understandable to store and inventory users.
- Validate the usefulness of suggested actions.
- Capture confirmed outcomes to support QA evidence and future model tuning.
- Identify false positives, false negatives and business rule refinements.

16.2 Validation Sample Design

Sample Segment	Recommended Sample	Purpose
Critical risk	Top 20–50 item/location records	Validate whether the highest scores are actionable and accurate
High risk	Random 20–50 records across categories/stores	Check model consistency beyond the top-ranked cases
Medium risk	10–20 records	Assess threshold calibration and early warning value
Low risk control group	10–20 records	Estimate false negatives and compare with high-risk population
Known historical issue cases	All available confirmed cases	Back-test whether the model would have flagged known phantom events

16.3 Business Validation Workflow

- Step 1: Generate daily or weekly risk output for the agreed pilot stores and categories.
- Step 2: Review top-ranked alerts with inventory control and store operations users.
- Step 3: Select validation sample based on risk band, category, store and commercial value.
- Step 4: Store users perform the recommended action, such as shelf check, backroom check, targeted count, receiving review or transfer review.
- Step 5: Capture validation outcome in the feedback table using standard reason codes.
- Step 6: QA compares captured outcomes with expected model behaviour and acceptance criteria.
- Step 7: Analytics team reviews false positives/false negatives and recommends threshold or formula tuning.

16.4 Validation Outcome Codes

Outcome Code	Description	Confirmed Phantom?
CONFIRMED_MISSING	Physical quantity is lower than system available stock and stock cannot be located	Yes
MISPLACED_FOUND	Stock exists but was not on shelf/selling location	Partial / operational issue
COUNT_VARIANCE	Targeted count confirms book-to-physical variance	Yes if physical < system
RECEIPT_ISSUE	Receiving documentation or physical receiving check identifies discrepancy	Yes/Root cause
TRANSFER_ISSUE	Transfer reconciliation identifies shipped vs received discrepancy	Yes/Root cause
DATA_OR_POS_ISSUE	Issue explained by sales transaction, data latency or audit exception	No, but data issue
STOCK_FOUND_CORRECT	Stock is physically present and available for sale	No
NO_ACTION	Alert was not validated	Unknown

16.5 Validation Metrics for QA and Business Review

Metric	Formula	Purpose
Action rate	Validated alerts / Alerts assigned for validation	Measures operational participation
Confirmation rate	Confirmed phantom alerts / Validated alerts	Measures precision of selected risk band
Critical precision	Confirmed critical alerts / Validated critical alerts	Checks whether highest scores are reliable
False positive rate	Stock found correct / Validated alerts	Identifies over-flagging
Misplaced stock rate	Misplaced found / Validated alerts	Quantifies on-shelf execution opportunity
Estimated value validated	Sum estimated lost sales value for confirmed alerts	Supports business case
Average time to validate	Average action_date - alert_date	Assesses operational feasibility
Explanation acceptance	Alerts where business agrees explanation is plausible / Validated alerts	Measures explainability quality

16.6 QA Evidence Pack

For each validation cycle, the QA team should retain:

- Input data snapshot used for scoring.
- Model version, parameters and risk weights applied.
- Generated risk output and driver output.
- Sample selection criteria.
- Business validation results and feedback records.
- Expected-result test cases for deterministic formula checks.
- Defects or discrepancies raised during validation.
- Approved tuning decisions and retest evidence.

17. QA Acceptance Criteria

Area	Acceptance Criteria
Eligibility	Inactive items, closed stores, non-ranged items and zero/negative available stock are excluded or flagged according to specification.
Formula reproducibility	QA can reproduce component risk scores for selected test records using documented formulas.
Risk score range	All component scores are between 0 and 1 and final risk score is between 0 and 100.
Risk bands	Risk bands are assigned according to the agreed thresholds.
Explanations	Every high and critical alert includes at least one meaningful explanation and top driver.
Output completeness	Mandatory output fields are populated for every generated alert.
Data lineage	Model version, business date and input snapshot timestamp are captured.
Feedback capture	Validation outcomes can be recorded and linked back to original alert_id.
Business validation	A defined sample is reviewed and outcomes are captured using standard codes.
Regression testing	Changes to weights or thresholds are retested against baseline scenarios.

18. Test Scenarios for QA

Scenario	Input Pattern	Expected Result
Eligible high-risk silent stock	Available stock > 0, no sales for many days, high expected demand	High or critical risk with silent stock as top driver
Slow-moving item	Available stock > 0, no sales, very low expected demand	Low or medium risk due to low demand confidence
Recent count variance	Recent physical count lower than book stock	High count variance risk and count-related explanation
Receipt discrepancy	Expected receipt quantity greater than received quantity, recent receipt	Receipt discrepancy risk and receiving investigation action

Transfer discrepancy	Transfer shipped quantity greater than received quantity	Transfer discrepancy risk and transfer reconciliation action
Inactive item	Item status inactive	Excluded from scoring
Closed store	Location status closed	Excluded from scoring
No available stock	Available stock <= 0	Excluded from phantom inventory scoring
Data freshness issue	Sales/stock snapshot not current	Record flagged or scoring batch fails according to severity
Explanation generation	High-risk record with multiple contributing drivers	Top drivers and explanation text populated

19. Implementation Phases

Phase	Activities	Deliverables
1. Data Discovery	Confirm MFCS/RAP data availability, profile required domains, agree available stock definition	Source mapping, data quality assessment, sample dataset
2. Feature Engineering	Build item/location/day dataset and risk features	Feature table, feature definitions, profiling report
3. Risk Engine	Implement formulas, weights, thresholds, confidence, priority and explanations	Risk result table, driver table, test results
4. Business Validation	Review top alerts, perform store checks/counts, capture outcomes	Validation report, confirmation metrics, tuning recommendations
5. QA and Stabilization	Execute functional, regression and data quality tests	QA evidence pack, defect log, sign-off recommendation
6. POC Playback	Present business case, demo ranked risks and explainability	POC playback deck, production roadmap

20. Configuration Parameters

Parameter	Recommended Default	Configurable By
Sales history window	56 comparable selling days	Category / department
Risk scoring window	14 days	Category / department
Demand confidence window	28 days	Global / category
Minimum demand threshold	3 units over 28 days	Category / department
Minimum movement days	7 days	Category / department
Count decay days	30 days	Global
Receipt decay days	21 days	Global
Transfer decay days	21 days	Global
Medium risk threshold	40	Global / pilot
High risk threshold	60	Global / pilot
Critical risk threshold	80	Global / pilot
Top-N alerts per store	To be agreed	Pilot governance

21. Future Enhancement Path

After the POC, the solution can evolve from explainable rules to a hybrid or supervised ML approach once enough validated labels are available.

```
Confirmed_Phantom_Inventory = 1 if physical count quantity < system available stock quantity by agreed tolerance
```

```
Future_Final_Risk_Score = α × Rule_Based_Risk_Score + (1 - α) × ML_Model_Risk_Score
```

Recommended future enhancements include supervised learning, root-cause classification models, operational task integration, replenishment unblock recommendations and closed-loop inventory accuracy reporting.

22. References and Assumptions

22.1 References

- Oracle Retail Merchandising Foundation Cloud Service documentation describes MFCS as supporting core merchandising activities including item management, inventory replenishment, purchasing, sales auditing and financial tracking.
- Oracle Retail AI Foundation / RAP documentation describes analytical and AI capabilities including advanced analytics and data science extensibility patterns.

22.2 Key Assumptions

- MFCS or its analytical replica (RAP) provides sufficient item/location stock and movement history for the selected pilot scope.
- Sales data is available at item/location/day level and is sufficiently cleansed for analytical use.
- AIF/RAP or another analytical platform can support the historical feature dataset.
- The POC external AI process will be read-only with respect to MFCS operational inventory.
- Business users will validate a controlled sample of alerts and provide structured feedback.
- The POC will use deterministic formula-based scoring (rules-based risk engine) before moving to supervised machine learning. The engine calculates phantom inventory risks using predefined business rules, normalized risk signals, configurable thresholds and weighted scoring formulas. This supports explainability, auditability and QA reproducibility before moving to supervised machine learning in a later phase.

23. Non-Functional Requirements

Area	Requirement
Batch Frequency	Daily
Runtime	<30 min
Explainability	Top 3 drivers
Auditability	Persist model version & snapshot
Security	Read-only MFCS access
Scalability	Configurable by store/category